trast to few-shot prompting, which gives instances and expects the model to produce comparable results, Chain of Thought prompting gives the model the steps it must take to find the answer. This approach enables the model to comprehend and interpret the critical stages needed to solve an issue better. The model is better able to reason and offer more precise answers when the issue is divided into smaller segments.

One such method is the Zero-Shot Chain of Thought prompting, which leads the model through a series of deductive steps to determine the right response. As shown in a study on zero-shot Chain of Thought prompting, this method may be utilised for specific activities and even functions without giving examples.

Large language models may execute more sophisticated reasoning tasks and perform better on a variety of arithmetic, commonsense, and symbolic reasoning problems when given a chain of reasoning.

The capacity of big language models to carry out complicated reasoning tasks is greatly enhanced by the generation of a chain of thought or a set of intermediate reasoning stages.

Example

The odd numbers in this group add up to an even number: 4, 8, 9, 15, 12, 2, 1.

A: Adding all the odd numbers (9, 15, 1) gives 25. The answer is False.

The odd numbers in this group add up to an even number: 17, 10, 19, 4, 8, 12, 24.

A: Adding all the odd numbers (17, 19) gives 36. The answer is True.

The odd numbers in this group add up to an even number: 16, 11, 14, 4, 8, 13, 24.

A: Adding all the odd numbers (11, 13) gives 24. The answer is True.

The odd numbers in this group add up to an even number: 17, 9, 10, 12, 13, 4, 2.

A: Adding all the odd numbers (17, 9, 13) gives 39. The answer is False.

The odd numbers in this group add up to an even number: 15, 32, 5, 13, 82, 7, 1.

A:

Output

Adding all the odd numbers (15, 5, 13, 7, 1) gives 41. The answer is False.

3. Selt Consistency Prompting

Self-consistency, as proposed by Wang et al. (2022), seeks "to replace the naive greedy decoding used in chain-of-thought prompting". The goal is to sample a variety of different lines of reasoning using few-shot CoT and then utilise the generations to choose the most reliable response. This improves the effectiveness of CoT prompting on tasks requiring numerical computation and common sense thinking.

Example:

When I was 6 my sister was half my age. Now I'm 70 how old is my sister?

Output:

35.

Now, clearly, this answer is not correct. Selt consistency prompting can help fix this issue. Here's how (as per an example shared by Wang, et al)

Q: There are 15 trees in the grove. Grove workers will plant trees in the grove today. After they are done,

there will be 21 trees. How many trees did the Grove workers plant today?

A: We start with 15 trees. Later we have 21 trees. The difference must be the number of trees they planted.

So, they must have planted 21 - 15 = 6 trees. The answer is 6.

Q: If there are 3 cars in the parking lot and 2 more cars arrive, how many cars are in the parking lot?

A: There are 3 cars in the parking lot already. 2 more arrive. Now there are 3 + 2 = 5 cars. The answer is 5.

Q: Leah had 32 chocolates and her sister had 42. If they ate 35, how many pieces do they have left in total?

A: Leah had 32 chocolates and Leah's sister had 42. That means there were originally 32 + 42 = 74

chocolates. 35 have been eaten. So in total, they still have 74 - 35 = 39 chocolates. The answer is 39.

Q: Jason had 20 lollipops. He gave Denny some lollipops. Now Jason has 12 lollipops. How many lollipops

did Jason give it to Denny?

A: Jason had 20 lollipops. Since he only has 12 now, he must